

Greening the Field

India's ambitious target of 500 GW non-fossil fuel capacity by 2030 will require extensive effort. Can it be achieved?

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FEBRUARY 7 WAS an important day in India's energy transition journey. The country achieved 100 GW installed solar capacity—a milestone on the path towards a greener future. This paves the way for the much bigger target of 500 GW non-fossil fuel capacity by 2030.

However, the road is long and mired in challenges. The electricity system in India faces the complex challenge of ensuring continuous and reliable power, while keeping costs affordable and reducing carbon emissions.

India's efforts to ramp up renewable energy (RE) capacity over

the last decade led to the country having the 4th largest installed capacity in 2023. It now has 211GW of RE installed capacity. By 2030, India is set to have the world's third-largest total installed renewable capacity, following only China and the US.

There is also a target of 15GW of installed nuclear power capacity by 2030 from the present 8GW. And efforts to reduce dependence on petroleum imports are apparent. For the first time, the Indian government has opened the nuclear sector to private participation; the companies can now invest in small and modular nuclear reactors.

The government's National Green Hydrogen Mission (NGHM)

